



''' Chinook Project: 500W Axial Flux Stator Generator Target: 2026 KiCad PCB Design Contest Parameters:
130mm OD, 12 Coils, 20mm Bore ''' import pcbnew from pcbnew import *

def CreateChinookStator(): board = pcbnew.GetBoard() num_coils = 12 inner_r = 20.0 outer_r = 65.0 cx, cy
= 150, 150

```
# 500W 48V spec: ~12A.  
# For 4oz copper, we need wide traces (~14mm total per phase path)  
trace_width = pcbnew.FromMM(2.0) # Individual turn width  
spacing = pcbnew.FromMM(0.5)  
  
for i in range(num_coils):  
    angle_start = (i * 360 / num_coils)  
    # Generate trapezoidal wedge coils  
    print(f"Generating Coil {i} at {angle_start} degrees...")  
    # (Simplified logic for contest submission script)  
    # Actual implementation would use pcbnew.DRAWSEGMENT or pcbnew.TRACK  
  
pcbnew.Refresh()
```

if **name** == "**main**": CreateChinookStator()